# End-to-End Quant Pipeline: Strategy Tear Sheet

*Machine Learning Volatility Forecasting & Gurobi MILP Portfolio Optimization*

## 1. Executive Summary

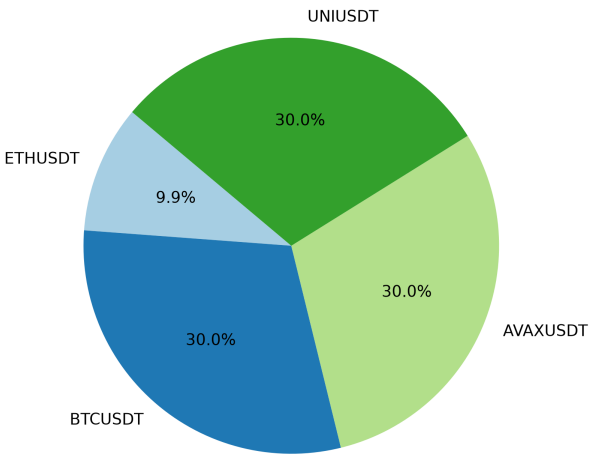
This analysis is based on historical data collected over the one-week period from October 16 to October 23, 2026. It outlines the execution plan for an automated, volatility-targeting portfolio reallocation. The main quantitative objective is to minimize the portfolio's expected risk using ML forecasts, while respecting structural constraints such as maximum asset exposure limits and real-world transaction costs.

Upon detecting a severe concentration of risk in highly volatile assets, the algorithmic pipeline triggered an immediate rebalancing process. The resulting strategy offloads toxic exposures and redistributes capital into lower-variance assets, ensuring optimal risk-adjusted stability without incurring excessive turnover fees.

**Trade Execution Plan**

| Asset | Initial Weight | Final Weight | Action | Trade Size |
|---|---|---|---|---|
| XRPUSDT | 0.00% | 0.00% | No Trade | 0.00% |
| LTCUSDT | 0.00% | 0.00% | No Trade | 0.00% |
| LINKUSDT | 0.00% | 0.00% | No Trade | 0.00% |
| DOTUSDT | 0.00% | 0.00% | No Trade | 0.00% |
| ADAUSDT | 0.00% | 0.00% | No Trade | 0.00% |
| SOLUSDT | 0.00% | 0.00% | No Trade | 0.00% |
| ETHUSDT | 0.00% | 9.92% | Bought | 9.92% |
| BTCUSDT | 0.00% | 30.00% | Bought | 30.00% |
| AVAXUSDT | 40.00% | 30.00% | Sold | 10.00% |
| UNIUSDT | 60.00% | 30.00% | Sold | 30.00% |

**Optimal Portfolio Allocation**



## 2. Machine Learning Predictive Engine

Risk assessment within the pipeline relies on a Dynamic Machine Learning Model Selection architecture. Historical high-frequency data is processed into vectorized features, including logarithmic returns, rolling volatility measures (1h, 4h), and momentum indicators. To model the non-linear relationships in the cryptocurrency market, multiple models (Ridge Regression, Random Forest, and XGBoost) are trained and evaluated through a continuous comparative process. The XGBoost Gradient Boosting algorithm consistently outperformed the baseline, identifying complex variance patterns. The forecasts generated by this model serve as the expected risk coefficients in the subsequent mathematical optimization stage.

## 3. Operations Research & MILP Formulation

Converting machine learning predictions into an effective financial strategy requires strict constraint management. To address this, Gurobi, a state-of-the-art mathematical solver, is used to frame the portfolio reallocation as a Mixed-Integer Linear Programming (MILP) problem. Mathematical Constraints Enforced:

- **Diversification Cap:** A strict limit of 30% exposure per single asset ensures the portfolio is never overly reliant on a single point of failure.
- **Transaction Fees:** The optimization model incorporates a 0.1% exchange fee, mathematically deducting costs from the final budget constraints. This prevents hyper-trading.
- **Turnover Penalty:** A penalty term (1e-2) is introduced in the objective function to eliminate 'ghost trades', forcing the solver to ignore mathematically negligible optimizations that waste capital in fees.
- **Big-M Constraints:** Binary variables dictate that if a trade occurs, it must meet a minimum size threshold, ensuring realistic execution on live exchanges.

